

Machine Learning Cycle

Objective:

You will demonstrate the end-to-end Machine Learning process.

Overview

As a Machine Learning engineer, you have been tasked with creating an ML Pipeline and scaling and monitoring it on a cloud platform of your choice. For this assignment, build on the use case for the introduction to Machine Learning Module Summative. **The extension on this particular assignment is to ensure you use data that is not tabular, preferably image or sound.**

Tasks

Creating a Machine Learning Classification model offline and deploying it.

Evaluate the model(s) using all the metrics required on a Jupyter Notebook and demonstrate how good the model(s) are.

The Breakdown:

Create the following processes:

- Data acquisition
- Data processing
- Model Creation
- Model testing

Model Retraining - in such a way that the model can be retrained again, and create a trigger for retraining the model when the need arises.

API creation with Python.

Create a UI to cover the following

Model up-time

Data Visualizations

Access to train and retrain functionalities

Deploy the processes in task 1 on a cloud platform. Demonstrate the evaluation process of the model in production.

Simulate a flood of requests (using software <https://locust.io/>) send them to the model, and show how the model responds to these requests. Record and show the latency and response time of the requests with different numbers of Docker containers.

Demonstrate how a user uploads values/features and the model predicts

A User should be able to upload new data and trigger retraining.

The final solution MUST have the following functionalities:

Model prediction - Allow a user to predict one datapoint from an image or sound

Visualizations - Create visualizations that make sense of different features in your dataset. Create interpretations of at least 3 features in your dataset. What story does it tell?

Upload Data - Bulk data that will be used to retrain (multiple images, multiple .wav files for sound)

Trigger retraining based on the uploaded data - Have a feature where one can press a button that can trigger a retraining process

Github Repo Directory Structure

```
Project_name/  
|  
├── README.md  
|  
└── notebook/
```

```

|   |—project_name.ipynb
|
|—src/
|   |—preprocessing.py
|   |—model.py
|   |—prediction.py
|
|—data/
|   |—train/
|   |—test/
|—models/
|   |—_model_name.pkl or _model_name.tf

```

Requirements

A link to the GitHub repo.

The README.md should have clear instructions on:

A video Demo - YouTube Link  YouTube

URL where applicable

The project description

And clear steps on how to set it up

Results from Flood Request Simulation 

Notebook

Should contain detailed preprocessing steps

Model Training

Model Test / Prediction Functions

The model file

Pickle (.pkl), tensorflow (.tf) or .h5 file

Submission Instructions

You will have two attempts during submission Make sure to submit the following in each attempt respectively

The first attempt will be a Zip File of the GitHub Code Repository

The second attempt will be a GitHub Repository URL

Rubric:

MLOPs

Criteria	How to	P
Video demo	5 to >3.0 ptsExcellent Clear and user-friendly demonstration of: The prediction and retraining process in a video with Camera on	5 pts
Retraining Process The script + Model file for retraining MUST be present to evaluate this criteria	10 to >7.5 ptsExcellent Clear Demonstration of the following triggers 1. Data file Uploading + Saving to Database (for purposes of retraining) 2. Data Preprocessing of the uploaded data 3. Retraining - The student uses a custom model created as a pre-trained model	10 pts

Prediction Process The script + model file for prediction MUST be present to evaluate this criterion	10 to >7.5 ptsExcellentClear Demonstration of the following: 1. Inserting a data point for prediction (Could be an image or inputs of a CSV row, or an audio file) 2. Displays the CORRECT prediction based on the label/class of the data point	10 pts
Evaluation of Models The notebook used to create the model MUST be present to evaluate this criteria	10 to >7.5 ptsExcellentClear Preprocessing steps are present with clear use of optimization techniques (Regularization, Optimizers, early stopping,use of a pretrained model or hyper parameter tuning) and Uses At least 4 Evaluation metrics used (Accuracy, loss, F1 score, Precision, Recall e.t.c)	10 pts
Deployment Package	10 to >7.0 ptsExcellentShowcases a UI using mobile app or Web app (Dockerized or public URL) Contains some data insights based on the dataset	10 pts